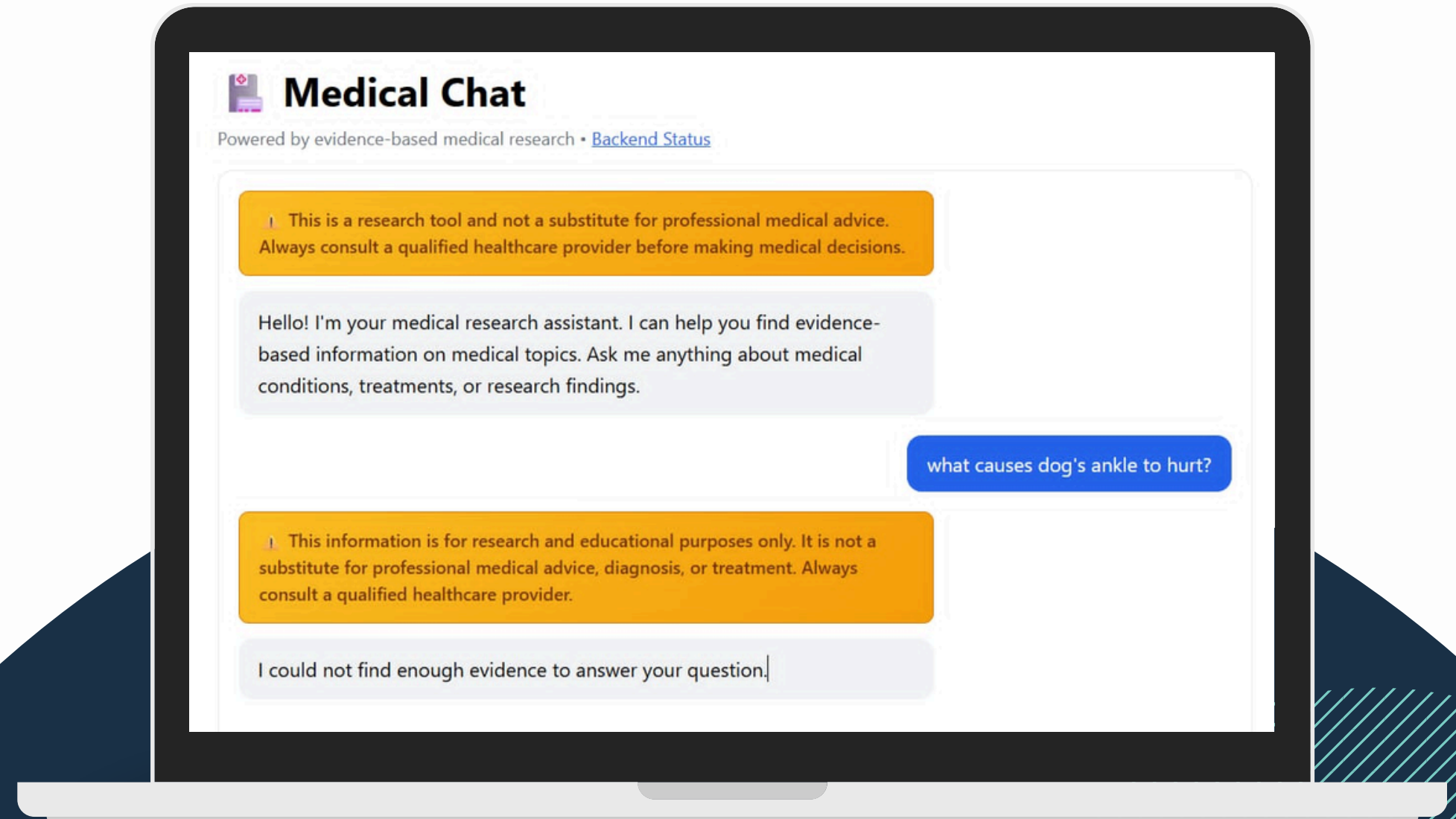
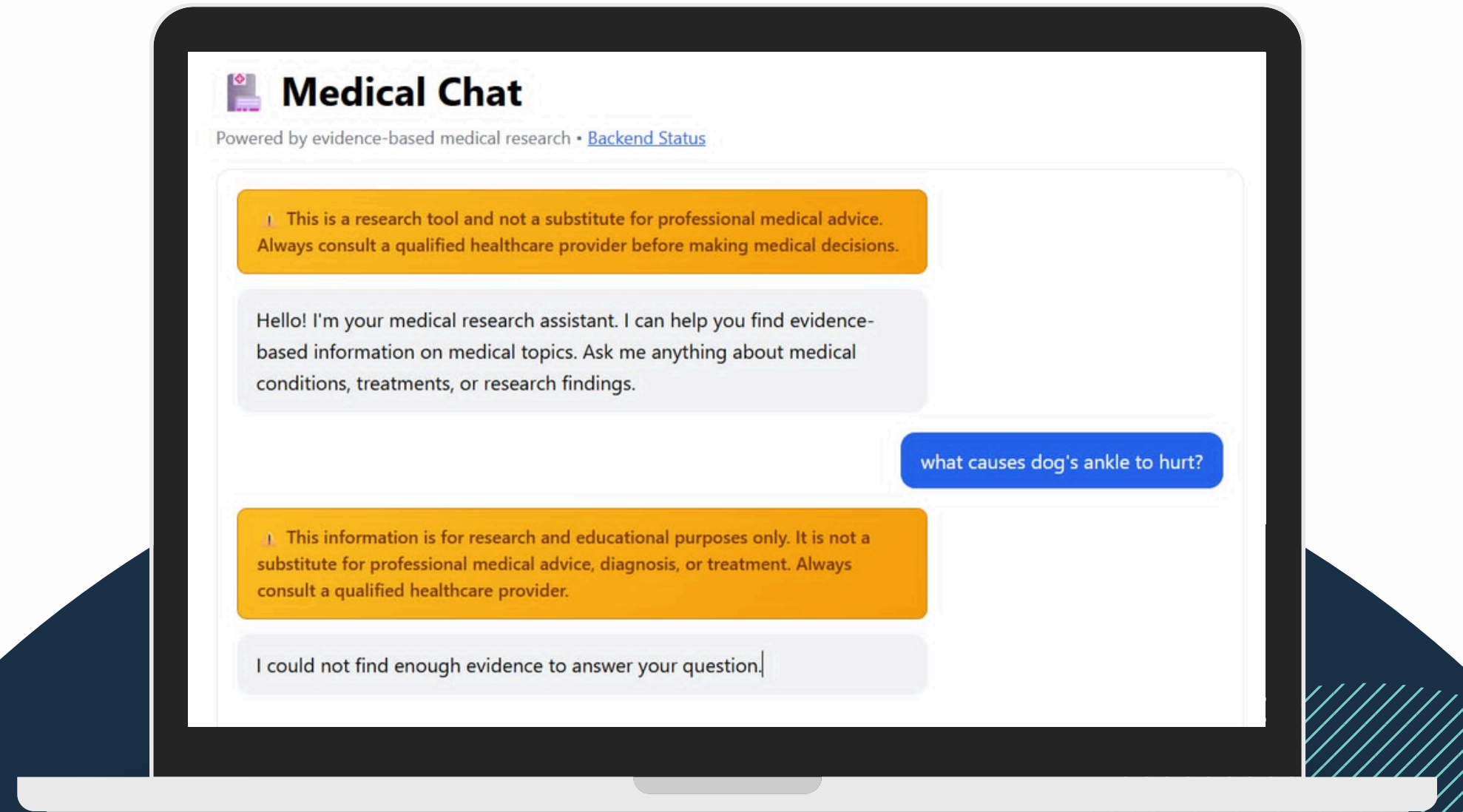


Medical Chat AI-Powered Medical Assistant



Github

<https://github.com/AntoninaGardzielewska/medical-chat>



Project overview

Medical Chat is an AI-powered medical chat assistant designed to provide evidence-based informational responses using:

- FastAPI (Python) backend
- Next.js (Node.js) frontend
- RAG (Retrieval-Augmented Generation) pipeline
- ChromaDB for vector search
- Ollama for local LLM inference

Key Idea

Build a conversational assistant for professionals that can answer medical questions using retrieved evidence from PubMed articles.

Important disclaimers: This project is an experimental proof of concept and does not provide medical advice.

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⚠ This is a research tool and not a substitute for professional medical advice. Always consult a qualified healthcare provider before making medical decisions.

Architecture

- **Frontend**

- Built with Next.js
- Provides the chat interface for users
- Communicates with the backend through REST API calls

- **Backend**

- Built with FastAPI
- Exposes the `/api/v1/chat` endpoint
- Implements the RAG pipeline:
 - Query rewriting
 - Vector retrieval
 - Answer synthesis with Ollama

- **Vector Database**

- ChromaDB stores embeddings of medical article chunks
- Enables semantic similarity search

- **LLM Layer**

- Ollama runs the language model locally
- Current default model: `llama3.2:latest`

- **Data Source**

- PubMed medical articles
- Retrieved via the NCBI E-utilities API

CI/CD

GitHub Actions runs on every push and pull request to main:

- Tests — pytest with coverage (Python 3.12 & 3.13)
- Style — Ruff lint + format check
- Types — mypy strict mode
- Security — Bandit SAST scan
- Coverage — uploaded to Codecov

Data Ingestion Pipeline

- ① Fetch PubMed articles.
- ② Chunk article text into smaller passages.
- ③ Generate embeddings for each chunk.
- ④ Store vectors and metadata in ChromaDB.

Key implementation details

- Implemented in `backend/ingestion/run_ingestion.py`.
- Creates the local database at `backend/chroma_db`.
- Populates the `pubmed_abstracts` collection.

How We Build the knowledge base

Step 1 - Fetch PubMed articles

Implemented in backend/ingestion/fetch.py.

Selection criteria

We query PubMed using predefined medical keywords: (because of our hardware limitations)

- “type 2 diabetes”
- “hypertension”
- “heart failure”

API parameters

- retmax=1000 → fetch up to 1000 results per term
- sort=relevance → PubMed returns the most relevant articles first
- Articles are retrieved in batches of 200 IDs

Filtering

We keep articles only if they contain:

- PMID
- Publication year
- Title
- Abstract text

Result

The script stores the selected articles in articles.json.

Number of articles: 2150

How We Build the knowledge base

Step 2 – Chunk the articles

Implemented in `backend/ingestion/chunk.py`.

Large articles are split into smaller semantic pieces so retrieval is more precise.

Chunking configuration

- Chunk size: 256 tokens
- Chunk overlap: 32 tokens
- Tokenizer: PubMedBERT embeddings tokenizer

Output

Each chunk contains:

- Text fragment
- Article metadata (title, authors, journal, year, PMID)
- Unique chunk ID

The chunks are saved to `chunked_articles.json`.

How We Build the knowledge base

```
=====
1. TOTAL CHUNKS IN DATABASE
=====
Total chunks stored: 7,034

=====
2. RANDOM SAMPLE (5 chunks)
=====

--- Chunk 1 ---
PMID      : 34251351
Title     : Type 2 Diabetes Mellitus and Cognitive Impairment.
Journal   : Acta medica Indonesiana
Year      : 2021
Text      : Type 2 diabetes mellitus (T2DM) is strongly associated with lower performance on multiple domains of cognitive function and with structural abnormalities of the brain. With the growing epidemic of dia...

--- Chunk 2 ---
PMID      : 35919809
Title     : Empagliflozin in the treatment of heart failure and type 2 diabetes mellitus: Evidence from several large clinical trials.
Journal   : International journal of medical sciences
Year      : 2022
Text      : Heart failure coexists with type 2 diabetes mellitus, which seriously affects the clinical treatment and prognosis. At present, the treatment for patients with established heart failure and type 2 dia...

--- Chunk 3 ---
PMID      : 30193140
Title     : Schizophrenia in type 2 diabetes mellitus: Prevalence and clinical characteristics.
Journal   : European psychiatry : the journal of the Association of European Psychiatrists
Year      : 2019
Text      : This study investigated the prevalence and characteristics of schizophrenia in patients with type 2 diabetes mellitus (T2DM) in Taiwan.

--- Chunk 4 ---
PMID      : 28322073
Title     : SGLT2 inhibitor/DPP-4 inhibitor combination therapy - complementary mechanisms of action for management of type 2 diabetes mellitus.
Journal   : Postgraduate medicine
Year      : 2017
Text      : empagliflozin and linagliptin or dapagliflozin and saxagliptin in the management of type 2 diabetes mellitus. Both combinations have been approved as single-pill formulations.

--- Chunk 5 ---
PMID      : 35056324
Title     : Impaired Folate-Mediated One-Carbon Metabolism in Type 2 Diabetes, Late-Onset Alzheimer's Disease and Long COVID.
Journal   : Medicina (Kaunas, Lithuania)
Year      : 2022
Text      : Impaired folate-mediated one-carbon metabolism (FOCM) is associated with many pathologies and developmental abnormalities. FOCM is a metabolic network of interdependent biosynthetic pathways that is k...
```

How We Build the knowledge base

Step 3 – Embed and store

Implemented in `backend/ingestion/embed_and_store.py`.

Embedding model

- PubMedBERT-based sentence embeddings
- Optimized for biomedical text

Storage

- Embeddings are inserted into ChromaDB
- Collection name: `pubmed_abstracts`
- Persistence path:
 - Local dev: `backend/chroma_db`
 - Docker: `/data/chroma_db`
- number of chunks: 7034

Runtime Query Pipeline (RAG)

What happens when a user asks a question?

Step 1 – User query

Example:

“What are the symptoms of heart failure?”

Step 2 – Retrieve relevant chunks

Implemented in backend/rag/retrieve.py.

Retrieval process

- The query is embedded into a vector.
- ChromaDB searches for semantically similar chunks.
- Results are filtered by similarity distance:
 - Only chunks with distance < max_distance are kept
 - The number of returned chunks is capped at max_results

Step 3 – Synthesize the answer

Implemented in backend/rag/synthesize.py and backend/rag/llm.py.

LLM workflow

- The retrieved chunks are combined into a prompt.
- The prompt is sent to Ollama via HTTP:
 - POST /api/generate
- Ollama generates a final natural-language answer grounded in the retrieved evidence.

Step 4 – Return response

The backend returns:

- generated answer
- optionally, metadata about supporting articles

How We Build the knowledge base

```
=====
```

3. SEMANTIC SEARCH TEST

```
=====
```

Query: 'treatment options for type 2 diabetes'

Result 1 (distance: 95.7119)

ID : 18514088_0

PMID : 18514088

Title : Intensifying treatment in poorly controlled type 2 diabetes mellitus: case reports.

Text : There is no single correct treatment option for all patients with type 2 diabetes mellitus. Duration of diabetes and stage of the disease, level of control, lifestyle habits, and attitude toward disea...

Result 2 (distance: 107.6119)

ID : 36650625_0

PMID : 36650625

Title : Evolution of Guideline Recommendations on Insulin Therapy in Type 2 Diabetes Mellitus Over the Last Two Decades: A Narrative Review.

Text : The prevalence of type 2 diabetes mellitus has been increasing worldwide. As the therapeutic options for type 2 diabetes mellitus have evolved over the last 2 decades, national and global guidelines r...

Result 3 (distance: 109.8022)

ID : 35259045_0

PMID : 35259045

Title : Cost-effectiveness of dapagliflozin compared to DPP-4 inhibitors as combination therapy with metformin in the treatment of type 2 diabetes mellitus without established cardiovascular disease in Colombia.

Text : SGLT2 inhibitors or DPP-4 inhibitors are among the preferred options in patients with type 2 diabetes mellitus (T2DM) without established cardiovascular disease.

Result 4 (distance: 109.8634)

ID : 29124701_0

PMID : 29124701

Title : Vitamin D and Type 2 Diabetes Mellitus.

Text : Type 2 diabetes mellitus (T2DM) has become a significant global health care problem and its reported incidence is increasing at an alarming rate. Despite the improvement in therapy and development of ...

Result 5 (distance: 111.7144)

ID : 21718606_0

PMID : 21718606

Title : Comprehensive treatment for type 2 diabetes mellitus.

Text : The treatment of type 2 diabetes mellitus (T2DM) is traditionally focused on glycemic control. In recent years, comprehensive treatment approaches including blood sugar control, blood pressure managem...

Example end-to-end flow

Example query

“What are common treatments for type 2 diabetes?”

System flow

- User sends question from the Next.js frontend.
- FastAPI receives the request at /api/v1/chat.
- Retriever searches ChromaDB for diabetes-related chunks.
- Top relevant chunks are selected and filtered by similarity distance.
- Ollama synthesizes an evidence-based answer using the retrieved context.
- Frontend displays the final response to the user.

Example answers

what are the possible causes of high blood sugar?

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Insulin resistance and super elevated blood sugar levels over a long time , impaired insulin release and sensitivity, elevated blood sugar, unfavourable changes in blood lipids

 Sources cited (2) ▶

Ask a medical question...

Send

what are the possible causes of high blood sugar?

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Insulin resistance and super elevated blood sugar levels over a long time , impaired insulin release and sensitivity, elevated blood sugar, unfavourable changes in blood lipids

 Sources cited (2) ▼

1. [1] [Diabetes mellitus and arterial hypertension. In search of the connecting link].

Bottermann P, Classen M

Zeitschrift fur die gesamte innere Medizin und ihre Grenzgebiete (1992)

[View on PubMed →](#)

2. [2] [Treatment of non-insulin dependent diabetes (type 2 diabetes mellitus)].

Hanssen K, Jenssen T

Tidsskrift for den Norske laegeforening : tidsskrift for praktisk medicin, ny raekke (1992)

[View on PubMed →](#)

Ask a medical question...

Send

Example answers



Medical Chat

Powered by evidence-based medical research • [Backend Status](#)

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Hello! I'm your medical research assistant. I can help you find evidence-based information on medical topics. Ask me anything about medical conditions, treatments, or research findings.

what causes dog's ankle to hurt?

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I could not find enough evidence to answer your question.

Ask a medical question...

Send

Thank you!

